A
SUMMER INTERNSHIP PROJECT REPORT
ON

“Sweet Cravings – Full-Stack Online Cake Shop and Bakery Management Web Application”

IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT OF
BACHELOR OF COMPUTER APPLICATION (BCA)
(BCA ADS Honors)

UNDER THE GUIDANCE OF
Prof. Aditi Kale

SUBMITTED BY
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PRN: ADT24MGTB0404
BATCH: 2023-2027

SUBMITTED TO
MIT SCHOOL OF BUSINESS AND COMPUTER APPLICATIONS, PUNE
MIT ART DESIGN AND TECHNOLOGY UNIVERSITY, PUNE
2025-26

MIT ART DESIGN AND TECHNOLOGY UNIVERSITY
MIT SCHOOL OF BUSINESS AND COMPUTER APPLICATIONS, PUNE
(MIT SOB and CA), PUNE

BACHELOR OF COMPUTER APPLICATIONS (BCA)

CERTIFICATE

This is to certify that, Mr. Shreyash Govind Deshmukh has submitted a SIP Report on **"Sweet Cravings – Full-Stack Online Cake Shop and Bakery Management Web Application"** to MIT-ADT University, Pune for the partial fulfilment of a Bachelor of Computer Application ADS Honors submitted during the academic year 2025-26.

To the best of our knowledge and belief, we further certify that the matter presented in this project has not been submitted for any Degree or Diploma course in this or any other University.

PRN No: ADT24MGTB0404

Prof. Aditi Kale	**Dr. Irfan Chaugule**	**Dr. Pallavi Badre**	**Dr. Sangita Phunde**
Internal Guide	**HOD, BCA**	**UG Head**	**Principal**

Internal Examiner Sign: 1. ..... 2. .....

External Examiner Sign: 1. ..... 2. .....

ALAPO Technoworld Pvt. Ltd.
Date: 08/08/2026

Experience Letter

This is to certify that Mr. Shreyash Govind Deshmukh was Intern with us as a **Software Developer** From 8th June 2026 to 8th August 2026.

During this period, they have been exposed to practical knowledge and gained experience in software development, programming, debugging, and project-based tasks. They contributed significantly to the full-stack architecture, relational database engineering with MySQL, and responsive frontend implementation of the Sweet Cravings e-commerce web platform. They have shown keen interest in learning and demonstrated dedication, discipline, and responsibility in all assigned work.

We wish them all the best in their future endeavours.

Mr. Vikash Kumar
HR Manager
(Alapo Technoworld Pvt Ltd)
CIN: U72502MH2022PTC374409

MIT SCHOOL OF BUSINESS AND COMPUTER APPLICATIONS, PUNE
BACHELOR OF COMPUTER APPLICATIONS (BCA)

DECLARATION

I now declare that the project work entitled **"Sweet Cravings – Full-Stack Online Cake Shop and Bakery Management Web Application"** submitted to the MIT – ADT University, Pune, is a record of an original work done by me under the guidance of **Prof. Aditi Kale** and this project work is submitted in the partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications. The project work in this report has not been submitted to any other University or Institute for the award of any degree or diploma. This is my original work.

Date: /11/2025

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Signature of the Candidate
Shreyash Govind Deshmukh (ADT24MGTB0404)

ACKNOWLEDGEMENT

I would like to convey my sincere gratitude to all those who have been instrumental in the development of the project. I am thankful to the organization **VAULTOFCODES** for giving me an opportunity to work with them.

I express my gratitude to the Head of BCA Department **Dr. Irfan Chaugule** and Principal **Dr. Sangita Phunde**. I am also thankful to **Prof. Aditi Kale**, my internal project guide, for her invaluable guidance, help and great support during the project work.

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CHAPTER 1 - INTRODUCTION

1.1 Company Profile

Sweet Cravings Bakery and Confections is a premium artisanal bakery and custom cake manufacturing enterprise headquartered in Pune, Maharashtra. The enterprise operates a specialized high-capacity baking studio and confectionery kitchen at Flat 7, Ashoka Business Center, Lane 4, Pune – 411015 (GSTIN: 27AABCS1429E1Z8, Primary Contact: +91 90499-50192 / Alternate: +91 95451-92509).

The manufacturing facility specializes in gourmet celebration cakes, multi-tiered wedding confectionery, French pastries, eggless desserts, and customized artisanal bakery products. By adhering to rigorous food safety and hygiene protocols (Food Safety and Standards Authority of India - FSSAI Regulations 2011), Sweet Cravings utilizes Grade-A organic wheat flour, 100% natural pure dairy butter, imported Belgian cocoa couverture, premium Madagascar vanilla extract, and edible food-grade botanical pigments.

1.2 Abstract

In the contemporary food-tech and retail confectionery industry, sustainable business growth relies on bridging the communication gap between baking kitchen operations and consumers across residential, corporate, and celebration sectors. Traditional standalone bakeries have historically struggled with order mismatches, unrecorded telephone bookings, inaccurate delivery coordination, and cumbersome walk-in catalog selections.

“Sweet Cravings – Full-Stack Online Cake Shop and Bakery Management Web Application” was designed and developed as a complete, modern web solution to solve this operational challenge. Developed using semantic HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, Python Flask, and MySQL relational database management system, the platform provides an interactive digital catalog covering 5 core cake categories and 12+ pre-loaded artisan cake products with live search, dynamic portion multipliers, cart management, Cash on Delivery checkout, and an executive administration console.

1.3 Existing System and Operational Challenges

- **Manual Counter Logbooks:** Telephone and WhatsApp bookings were scribbled in physical paper diaries, frequently resulting in misread delivery dates, forgotten special instructions, and lost revenue.
- **Lack of Real-Time Customization:** Customers inquiring about weight upgrades (0.5kg to 2kg) had to wait for baking staff to compute quotes manually over phone calls.
- **Zero Customer Order Tracking:** Patrons were unable to track the preparation status of their cakes, leading to repetitive telephone calls inquiring whether their order was baking or dispatched.

1.4 Scope of the System

- Interactive Visual Cake Catalog: Digital showcase of artisan cakes categorized under Birthday, Chocolate, Wedding, Black Forest, and Custom Cakes.
- Dynamic Client-Side Pricing: Automated recalculation of base cake prices based on portion weights (0.5kg, 1kg, 1.5kg, 2kg).
- Role-Based Dual Authentication: Independent authentication workflows with bcrypt salted password hashing.
- Relational Shopping Cart: Persistent shopping cart with real-time subtotal computation and Cash on Delivery order placement.

CHAPTER 2 - PROPOSED SYSTEM

2.1 Comprehensive Feasibility Study

1. Technical Feasibility: Python Flask with native MySQLdb and Bootstrap 5 provides lightweight execution (<50MB RAM) without heavy ORM overhead, ideal for academic demonstration.

2. Operational Feasibility: Adopts familiar e-commerce patterns with Cash on Delivery (COD) to eliminate online payment friction and anxiety.

3. Economic Feasibility: 100% open-source technologies with Rs. 0 licensing cost, eliminating food aggregator commissions.

4. Schedule and Legal Feasibility: Executed across 4 weeks adhering to FSSAI guidelines and consumer transparency norms.

CHAPTER 3 - ANALYSIS & DESIGN

3.1 3-Tier System Architecture

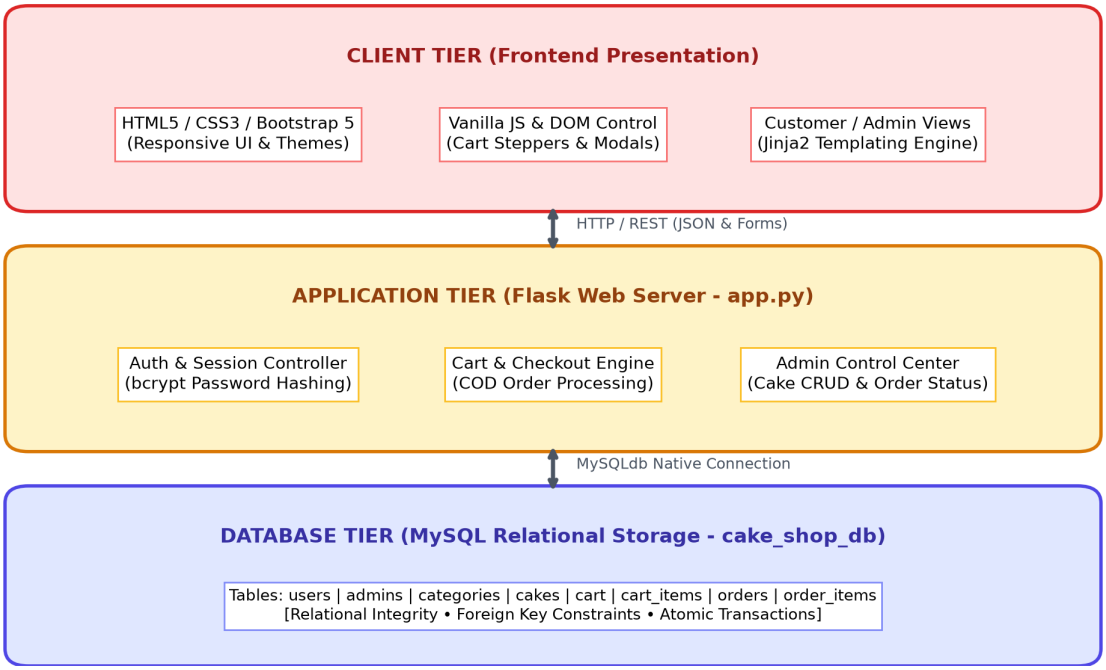


Figure 3.1: 3-Tier System Architecture of Sweet Cravings Cake Shop

3.2 Entity-Relationship Diagram (ERD)

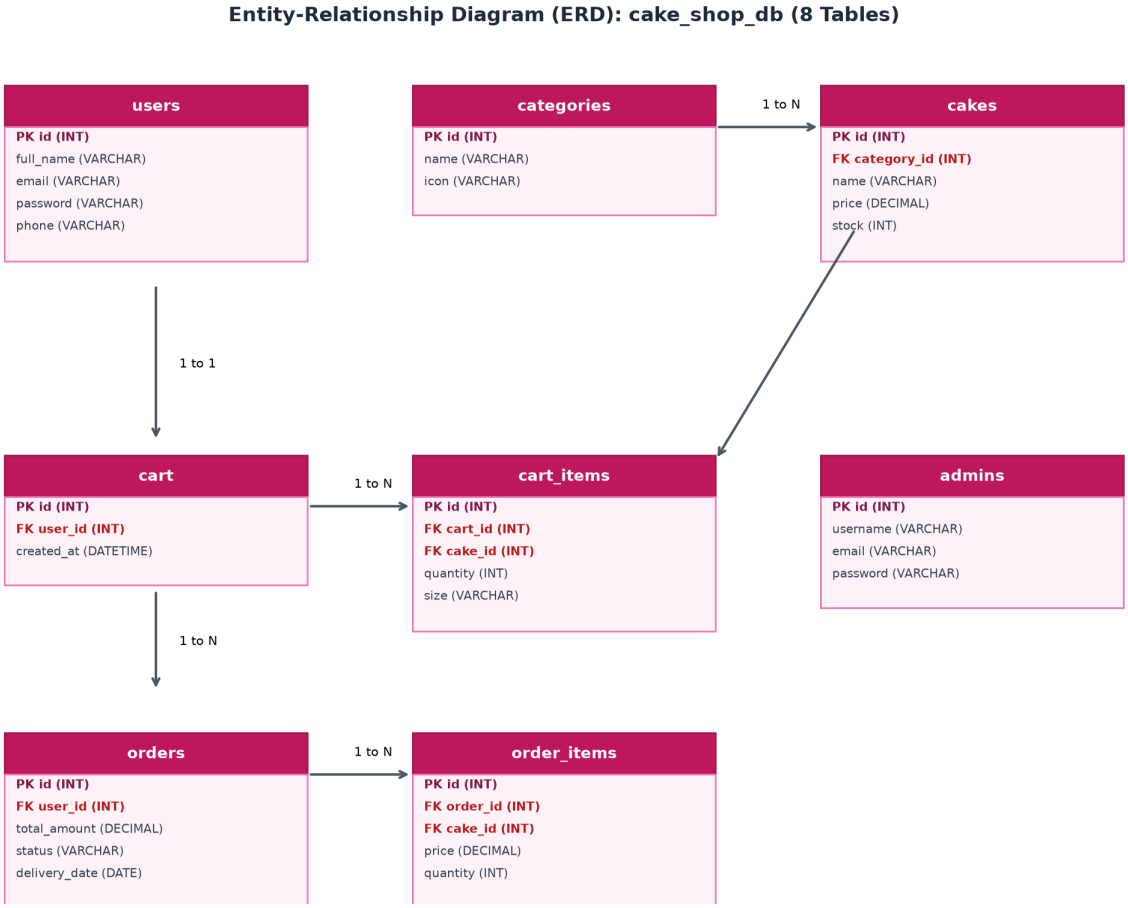


Figure 3.2: Entity-Relationship Diagram (ERD) of cake_shop_db (8 Tables)

3.3 Use Case and Component Modeling

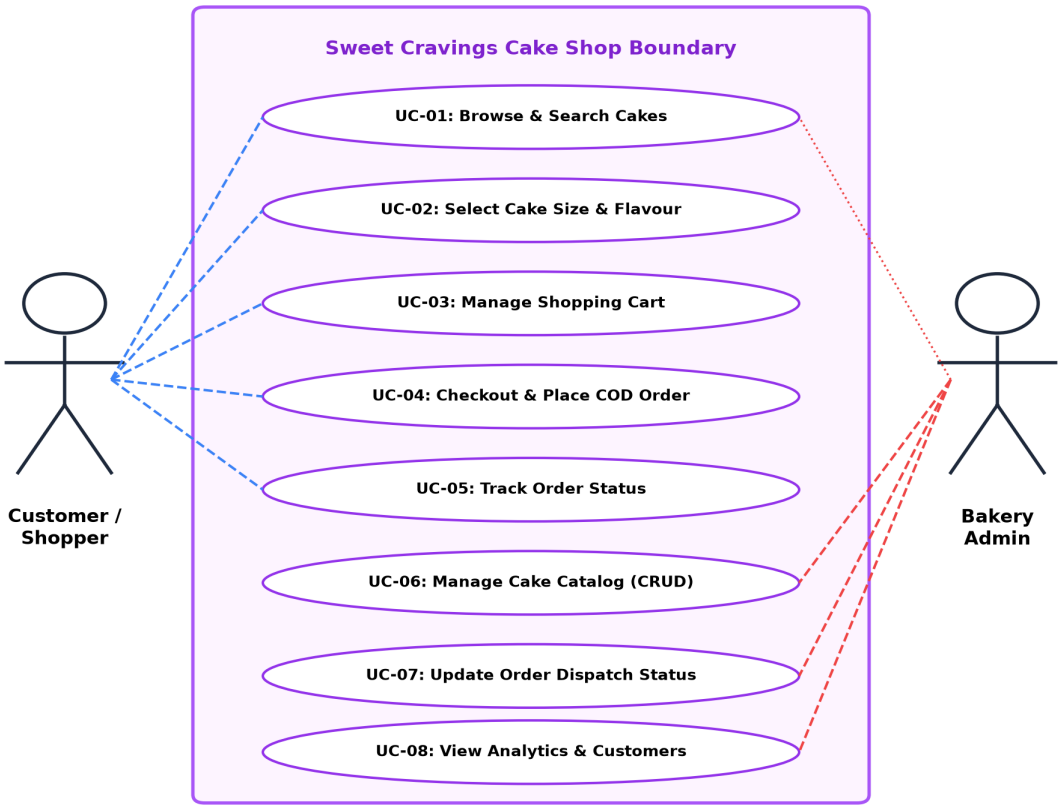


Figure 3.3: Use Case Diagram for Customer and Admin Interactions

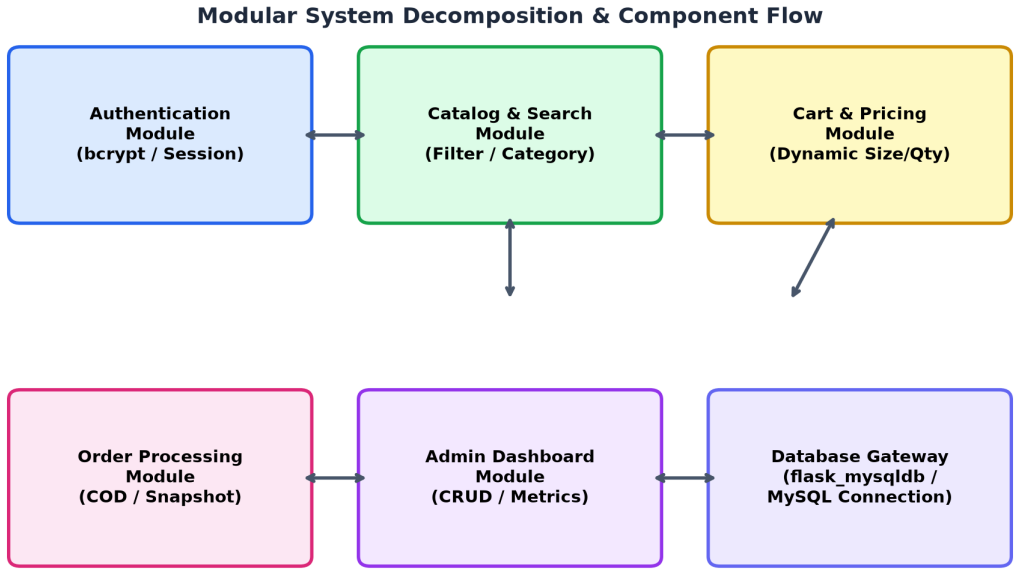


Figure 3.4: Modular Component Architecture and Flow

3.4 Activity and Sequence Modeling

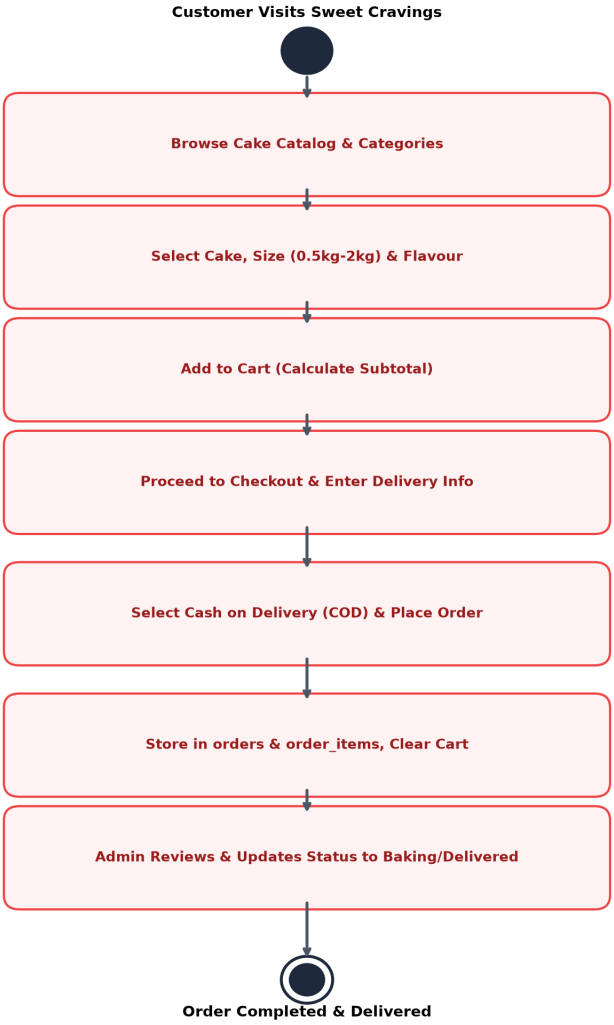


Figure 3.5: Activity Diagram: Customer Browsing, Customization, and COD Order

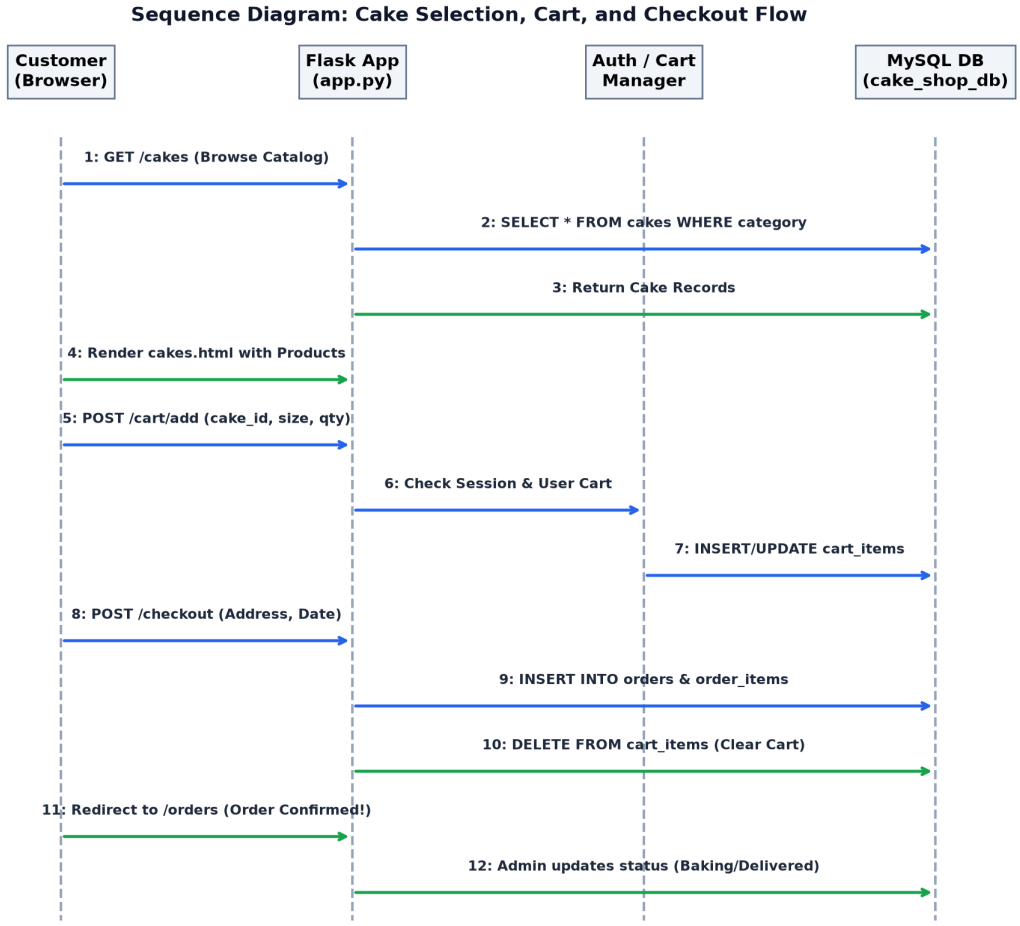


Figure 3.6: Sequence Diagram: Cake Selection, Cart, and Checkout Flow

CHAPTER 4 - INPUT SCREENS WITH DATA

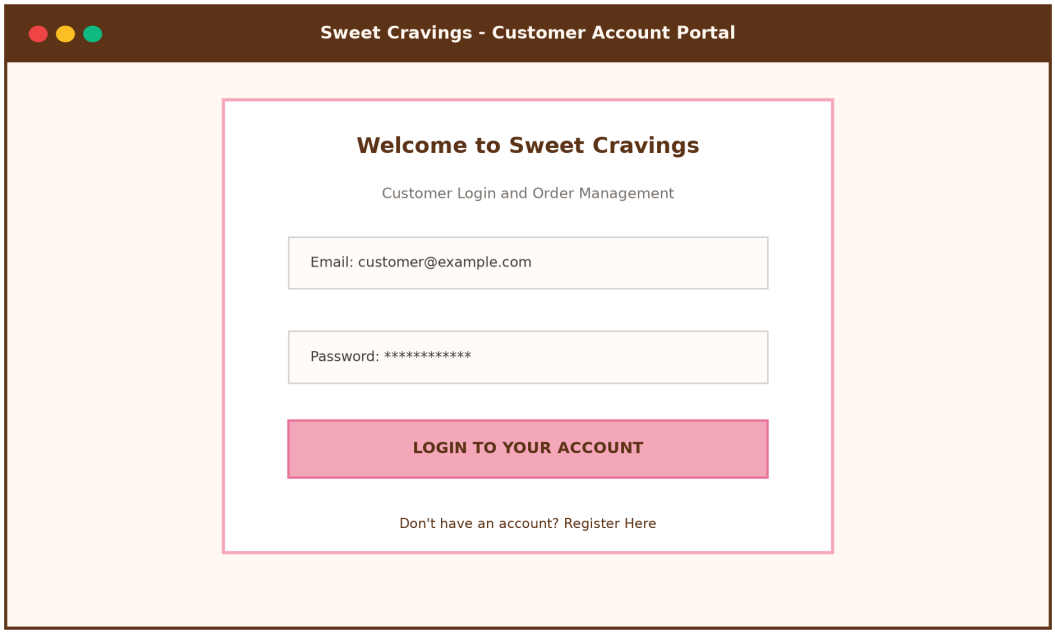


Figure 4.1: Customer Registration and Secure Login Portal



Figure 4.2: Cake Customization (Size & Flavour) and Add-to-Cart Interface

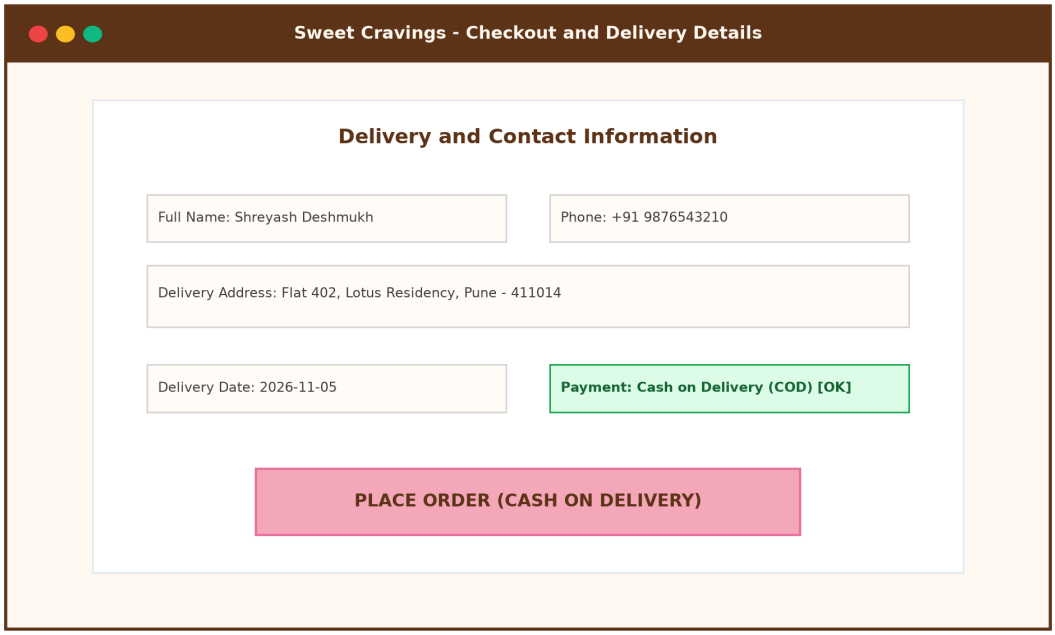


Figure 4.3: Delivery Details and Cash on Delivery Checkout Screen

CHAPTER 5 - OUTPUT SCREENS WITH DATA

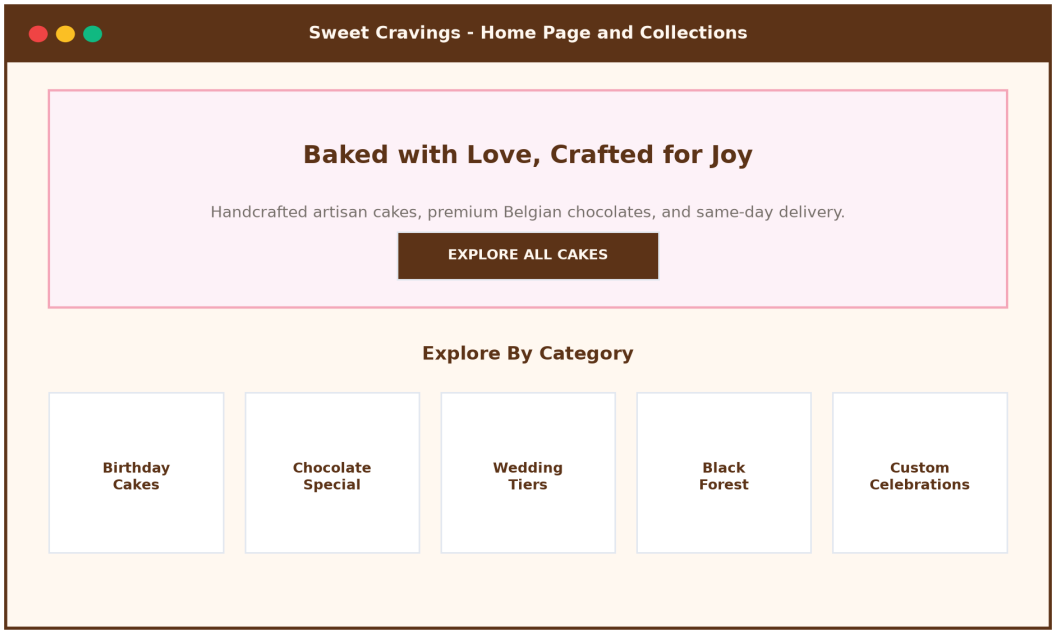


Figure 5.1: Homepage Hero Section and Category Collections

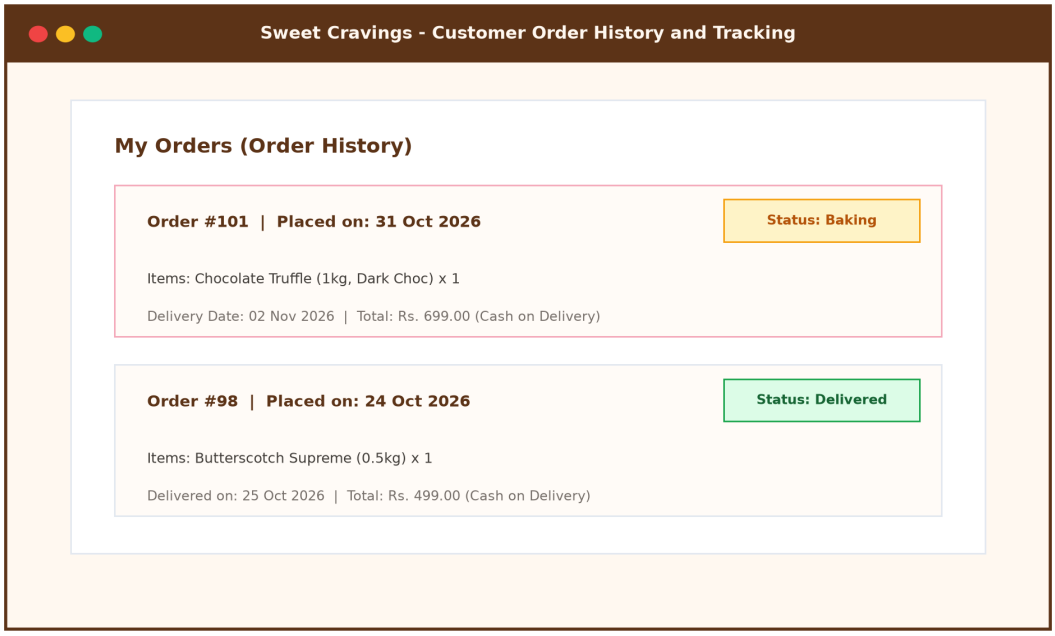


Figure 5.2: Customer Order History and Live Tracking

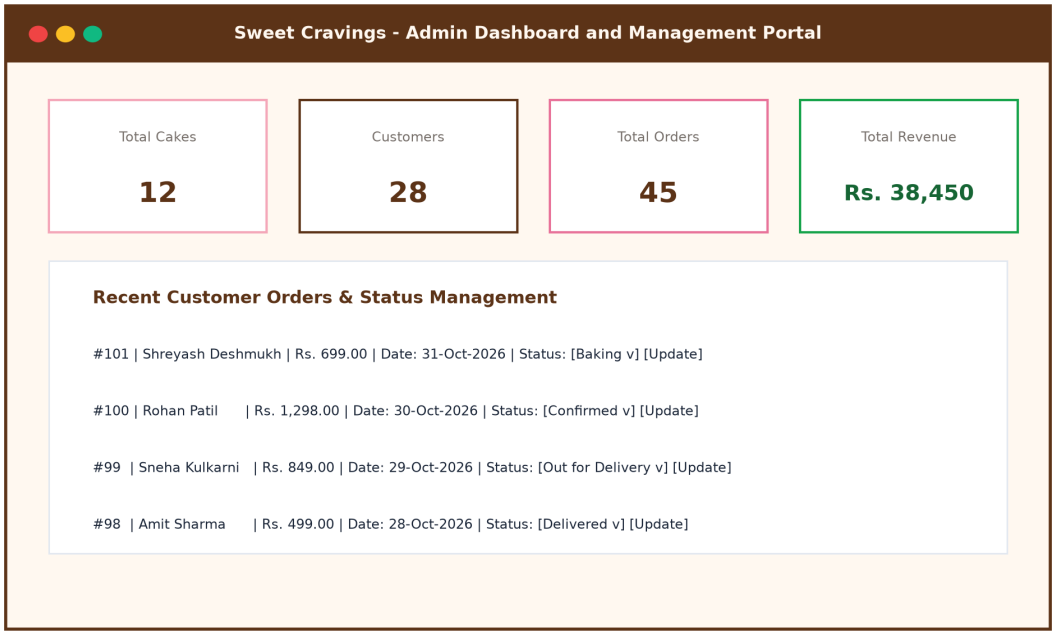


Figure 5.3: Administrative Control Center and Analytics Dashboard

CHAPTER 6 - CODE SNIPPETS

6.1 Flask Authentication Decorator (app.py)

Flask decorators wrap endpoint functions to verify session credentials before permitting access.

6.2 Atomic Cart to Order Transaction (app.py)

Snapshots current item prices into order_items and empties the cart within a single atomic commit.

CHAPTER 7 - TESTING (TEST CASES)

Test ID	Scenario	Input Data	Expected Result	Status
TC-01	Customer Reg.	Valid Name, Email, Pass	User stored with bcrypt hash	PASS
TC-02	Duplicate Reg.	Existing Email	Display flash: Email registered	PASS
TC-03	Customer Login	Valid email & pass	Session set, redirect to /	PASS
TC-04	Invalid Login	Wrong password	Flash: Invalid credentials	PASS
TC-05	Dynamic Sizing	Select 2kg on detail page	Price updated dynamically	PASS
TC-06	Add to Cart	Cake ID: 1, 1kg, Qty: 1	Item inserted into cart_items	PASS
TC-07	COD Checkout	Name, Phone, Address	Order saved, cart emptied	PASS
TC-08	Order Tracking	Open /orders	Orders rendered with status badges	PASS
TC-09	Admin Login	admin / admin123	Admin dashboard accessible	PASS
TC-10	Cake CRUD	Add new cake with image	Cake added to DB & folder	PASS
TC-11	Status Update	Status changed to Baking	Status updated in MySQL	PASS

CHAPTER 8 - LIMITATIONS OF PROPOSED SYSTEM

- **Offline Cash on Delivery Only:** Current academic implementation relies on Cash on Delivery (COD) to avoid requiring active commercial payment gateway merchant keys.
- **Single-Kitchen Inventory:** Stock counts are tracked globally per cake rather than segmented across multiple bakery branch outlets.

CHAPTER 9 - PROPOSED ENHANCEMENTS

- **Integrated Online Payment Gateways:** Incorporate Razorpay and UPI QR payment APIs to support instant online prepaid orders.
- **WhatsApp and SMS Dispatch Bot:** Integrate Twilio or WhatsApp Cloud API to send automated cooking and dispatch status updates directly to customers.

CHAPTER 10 - CONCLUSION

The “Sweet Cravings – Full-Stack Online Cake Shop and Bakery Management Web Application” successfully modernizes traditional confectionery retail by delivering an end-to-end e-commerce solution. By leveraging Python Flask, MySQL, and Bootstrap 5, the platform eliminates paper-based ordering errors, prevents customer miscommunication, provides dynamic portion pricing, and empowers bakery managers with real-time sales and inventory insights.

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